Started on	Tuesday, 28 April 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 28 April 2026, 11:50 AM
Time taken	10 mins 1 sec

Question 1
Complete
Marked out of 2.00

Let a,b,c be real numbers such that both b and c are not equal to 0.

Suppose P = [a b
c 0]
and P^{-1} = P. Then which of the following is true

- a. a = 0 and bc = -1
- b. a = 0 and bc =1
- c. a is not equal to 0 and bc = 1
- d. a = 0 and bc = 2

Question 2
Complete
Marked out of 2.00

If A is a 3 × 5 matrix, then the rank of transpose of A is at most

- a. 4
- b. 2
- c. 3
- d. 5

Question 3

Complete

Marked out of 2.00

Find the wrong one from the given statements

- a. If A is diagonalizable and invertible, then A^{-1} is diagonalizable
- b. None of the above
- c. If A is diagonalizable, then transpose of A is also diagonalizable
- d. An n × n matrix with fewer than n distinct eigenvalues is not diagonalizable

Question 4

Complete

Marked out of 2.00

Let P be a 4x4 matrix comprising of the following rows:

[1 2 0 2] , [-1 -2 1 1], [1 2 -3 -7], [1 2 -2 -4]

Then rank of P equals

- a. 3
- b. 1
- c. 4
- d. 2

Question 5

Complete

Marked out of 2.00

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

- a. 1U/144
- b. 125/432
- c. 325/432
- d. 13//144

Question 6

Complete

Marked out of 2.00

Let E and F be two events with 0 < P(E) < 1, 0 < P(F) < 1 and P(E|F) > P(E) . Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

- a. E and F are independent
- b. P(F | E^C) < P(F)
- c. P(F|E) > P(F)
- d. P(E | F^C) > P(E)

Question 7

Complete

Marked out of 2.00

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/U and P(F) < P(E). Then P(E) equals

- a. 3/4
- b. 2/3
- c. 1/3
- d. 1/2

Question 8

Complete

Marked out of 2.00

Politicians are certainly interested in knowing what fraction of the voters will favor them in the next election. which test will be used for this purpose given to you a sample with voters and nonvoters.

- a. z-test
- b. f-test
- c. chisquare test
- d. t-test

Question U

Complete

Marked out of 2.00

Blood Sample Data: In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. fi. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. fi second blood sample was obtained from each deer 30 minutes after the first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given. fissuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes. Which test to use

a. paired chisquare test

b. paired f test

c. paired z test

d. paired t test

Question 10

Not answered

Marked out of 2.00

An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. how many degree of freedom

a. 18

b. 10

c. 15

d. 20

◄ OS CSE

Jump to...

SMCS-CSE ►

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Completed on	Tuesday, 28 April 2026, 11:49 AM
Time taken	9 mins 52 secs

Question **1**

Complete

Marked out of 2.00

If A is a 3 × 5 matrix, then the rank of transpose of A is at most

- [x] a. 5
- [ ] b. 4
- [ ] c. 3
- [ ] d. 2

Question **2**

Complete

Marked out of 2.00

Pick the odd one out

- [ ] a. There is a nonzero vector x such that $Ax = \lambda x$
- [x] b. The system of equations $(\lambda I - A)x = 0$ has trivial solutions.
- [ ] c. λ is an eigenvalue of A
- [x] d. λ is a solution of the characteristic equation $\det(\lambda I - A) = 0$.

Question **3**

Complete

Marked out of 2.00

A square matrix A is invertible if and only if

- [ ] a. 0 is an eigenvalue of A
- [ ] b. 1 is an eigenvalue of A
- [ ] c. 1 is not an eigenvalue of A
- [x] d. 0 is not an eigenvalue of A

Question **4**

Complete

Marked out of 2.00

If A is as given, then A^{-1} can be given as

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

a. $\frac{1}{4}((A^2+6A+9\mathbf{I}))$

b. $\frac{1}{4}((A^2-6A-9\mathbf{I}))$

c. None of the above

d. $\frac{1}{4}((A^2-6A+9\mathbf{I}))$

Question **5**

Complete

Marked out of 2.00

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

a. 0.029

b. 0.039

c. 0.034

d. 0.044

Question **6**

Complete

Marked out of 2.00

Let E and F be two events with $0 < P(E) < 1$, $0 < P(F) < 1$ and $P(E|F) > P(E)$. Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

- [ ] a. $P(F|E) > P(F)$
- [x] b. E and F are independent
- [x] c. $P(F \mid E^C) < P(F)$
- [ ] d. $P(E \mid F^C) > P(E)$

Question **7**

Complete

Marked out of 2.00

If X

is a continuous random variable with PDF $f(x) = k x^2$

for $0 <= x <= 1$, the value of k

is:

- [x] a. 3
- [ ] b. 2
- [ ] c. 1
- [ ] d. 4

Question **8**

Complete

Marked out of 2.00

For comparing the means of weight in the two groups -smokers and non-smoker, which visualization is appropriate?

- ○ a. pie chart
- ○ b. boxplot
- ○ c. line graph
- ◉ d. scatterplot

Question **9**

Complete

Marked out of 2.00

The

_ _______ of a test is the probability of rejecting H0 given that a specific alter-
native is true.

- a. tensor
- b. power
- c. significance
- d. strength

Question 10

Complete

Marked out of 2.00

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

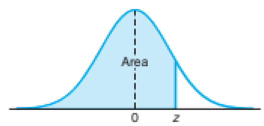
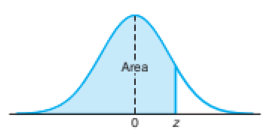


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183



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−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
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−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

a. +- 2.56

b. +-2.22

c. +-2.96

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Question **1**

Complete

Marked out of 2.00

The rank of matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$ is

- a. 1
- b. max(2,3)
- c. min(2,3)
- d. None of the above

Question **2**

Complete

Marked out of 2.00

$$A = \begin{pmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{pmatrix}$$

If A is as given the for what values of a does the system AX=Y has unique solution for any Y

- a. None of the above
- b. Both
- c. -2
- d. 1

Question **3**

Complete

Marked out of 2.00

If A is 3x3 matrix with all real entries, has an eigenvalue i

, then its other two eigenvalues can be

- a. -1,i
- b. 0,-i
- c. 2i,-2i
- d. 0,1

Question **4**

Complete

Marked out of 2.00

If A is a 3×5 matrix, then the rank of transpose of A is at most

- a. 5
- b. 3
- c. 4
- d. 2

Question **5**

Complete

Marked out of 2.00

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

- a. 0.044
- b. 0.039
- c. 0.029
- d. 0.034

Question **6**

Complete

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Question **7**

Complete

Marked out of 2.00

If X

is a continuous random variable with PDF $f(x) = k\,x^2$

for $0 <= x <= 1$, the value of k

is:

- [ ] a. 4
- [x] b. 3
- [ ] c. 1
- [ ] d. 2

Question **8**

Complete

Marked out of 2.00

For comparing the means of weight in the two groups -smokers and non-smoker, which visualization is appropriate?

- ( ) a. pie chart
- ( ) b. line graph
- ( ) c. boxplot
- (•) d. scatterplot

Question **9**

Complete

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A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

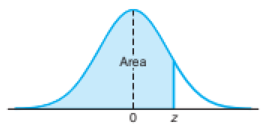

Table A.3 Areas under the Normal Curve

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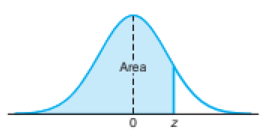

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−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
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- a. +-2.96
- b. +-2.22
- c. +- 2.56

Question **10**

Complete

Marked out of 2.00

The more prevalent situations involving tests on two means are those in which variances are unknown. If the scientist involved is willing to assume that both distributions are normal and that $\sigma 1 = \sigma 2 = \sigma$, the test used is

- a. one sample t-test
- b. cross-origin t-test
- c. variational t-test
- d. marginal t-test
- e. pooled t-test

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Started on	Tuesday, 28 April 2026, 11:40 AM
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Question **1**
Complete
Marked out of 2.00

If A is a 3 × 5 matrix, then the rank of transpose of A is at most

- [x] a. 5
- [ ] b. 3
- [ ] c. 2
- [ ] d. 4

Question **2**
Complete
Marked out of 2.00

Suppose that the characteristic polynomial of some matrix A is found to be
p(λ) = (λ − 1)(λ − 3)^2 (λ − 4)^3 . What is the size of A?

- [x] a. 6 X 5
- [ ] b. 5 X 5
- [x] c. 6 X 6
- [ ] d. 5 X 6

Question **3**
Complete
Marked out of 2.00

Let a,b,c be real numbers such that both b and c are not equal to 0.

Suppose P = [a b
c 0]
and P^{-1} = P. Then which of the following is true

- [ ] a. a is not equal to 0 and bc = 1
- [ ] b. a = 0 and bc = 2
- [x] c. a = 0 and bc =1
- [x] d. a = 0 and bc = -1

Question **4**

Complete

Marked out of 2.00

Find the wrong one from the given statements

- a. An n × n matrix with fewer than n distinct eigenvalues is not diagonalizable
- b. None of the above
- c. If A is diagonalizable and invertible, then A^{-1} is diagonalizable
- d. If A is diagonalizable, then transpose of A is also diagonalizable

Question **5**

Complete

Marked out of 2.00

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

- a. 0.044
- b. 0.039
- c. 0.034
- d. 0.029

Question **6**

Complete

Marked out of 2.00

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

- a. 325/432
- b. 13//144
- c. 19/144
- d. 125/432

Question **7**
Complete
Marked out of 2.00

The chances of failing in Physics are 20%
and in Mathematics are 10%

The chance of failing in at least one subject is

- [ ] a. 58%
- [ ] b. 28%
- [x] c. 38%
- [ ] d. 62%

Question **8**
Complete
Marked out of 2.00

The more prevalent situations involving tests on two means are those in which variances are unknown. If the scientist involved is willing to assume that both distributions are normal and that σ1 = σ2 = σ, the test used is

- ○ a. marginal t-test
- ○ b. one sample t-test
- ○ c. cross-origin t-test
- ● d. pooled t-test
- ○ e. variational t-test

Question **9**
Complete
Marked out of 2.00

Blood Sample Data: In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given. Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes. Which test to use

- ● a. paired t test
- ○ b. paired chisquare test
- ○ c. paired z test
- ○ d. paired f test

Question **10**

Complete

Marked out of 2.00

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

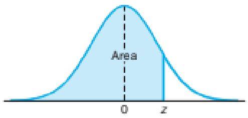

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

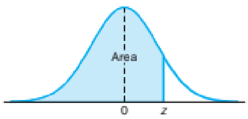

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
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−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

- a. +-2.22
- b. +- 2.56
- c. +-2.96

◀ CCN CSE

Jump to...

SMCS-CSE ▶

Started on	Tuesday, 28 April 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 28 April 2026, 11:49 AM
Time taken	9 mins 22 secs

Question **1**
Complete
Marked out of 2.00

Let P be a 4x4 matrix comprising of the following rows:

[1 2 0 2] , [-1 -2 1 1], [1 2 -3 -7], [1 2 -2 -4]

Then rank of P equals

- [x] a. 1
- [ ] b. 4
- [ ] c. 3
- [ ] d. 2

Question **2**
Complete
Marked out of 2.00

A square matrix A is invertible if and only if

- [ ] a. 1 is an eigenvalue of A
- [ ] b. 0 is an eigenvalue of A
- [x] c. 0 is not an eigenvalue of A
- [ ] d. 1 is not an eigenvalue of A

Question **3**
Complete
Marked out of 2.00

Suppose that the characteristic polynomial of some matrix A is found to be

p(λ) = (λ − 1)(λ − 3)^2 (λ − 4)^3 . What is the size of A?

- [ ] a. 5 X 5
- [ ] b. 5 X 6
- [x] c. 6 X 6
- [ ] d. 6 X 5

Question **4**

Complete

Marked out of 2.00

If A is a 3 × 5 matrix, then the rank of transpose of A is at most

- [x] a. 5
- [ ] b. 3
- [ ] c. 4
- [x] d. 2

Question **5**

Complete

Marked out of 2.00

The chances of failing in Physics are 20%

and in Mathematics are 10%

The chance of failing in at least one subject is

- [ ] a. 62%
- [ ] b. 38%
- [x] c. 28%
- [ ] d. 58%

Question **6**

Complete

Marked out of 2.00

Let E and F be two events with 0 < P(E) < 1, 0 < P(F) < 1 and P(E|F) > P(E) . Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

- [ ] a. E and F are independent
- [ ] b. P(F|E) > P(F)
- [x] c. P(F | E^C) < P(F)
- [ ] d. P(E | F^C) > P(E)

Question **7**

Complete

Marked out of 2.00

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

- [ ] a. 0.034
- [ ] b. 0.039
- [x] c. 0.044
- [ ] d. 0.029

Question **8**

Complete

Marked out of 2.00

For comparing the means of weight in the two groups -smokers and non-smoker, which visualization is appropriate?

- ( ) a. line graph
- (•) b. scatterplot
- ( ) c. boxplot
- ( ) d. pie chart

Question **9**

Complete

Marked out of 2.00

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

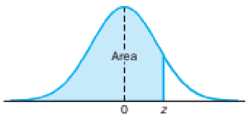

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
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−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
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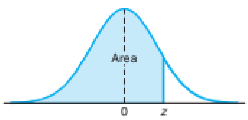

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−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

- a. +-2.96
- b. +- 2.56
- c. +-2.22

Question **10**

Complete

Marked out of 2.00

A ____________is the lowest level (of significance) at which the observed value of the test statistic is significant.

- a. alpha
- b. p value
- c. z value
- d. beta
- e. f value

Started on	Tuesday, 28 April 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 28 April 2026, 11:49 AM
Time taken	9 mins 39 secs

Question **1**
Complete
Marked out of 2.00

Suppose that the characteristic polynomial of some matrix A is found to be
p(λ) = (λ − 1)(λ − 3)^2 (λ − 4)^3 . What is the size of A?

- a. 6 X 5
- b. 5 X 5
- c. 6 X 6
- d. 5 X 6

Question **2**
Complete
Marked out of 2.00

The rank of matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$ is

- a. min(2,3)
- b. None of the above
- c. max(2,3)
- d. 1

Question **3**

Complete

Marked out of 2.00

$$A = \begin{pmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{pmatrix}$$

If A is as given the for what values of a does the system AX=Y has unique solution for any Y

- a. 1
- b. -2
- c. None of the above
- d. Both

Question **4**

Complete

Marked out of 2.00

If A is as given, then find sum of eigenvalues of A^{-1}

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

- a. 12/19
- b. None of the above
- c. 19/12
- d. 8

Question **5**

Complete

Marked out of 2.00

Let E and F be two events with 0 < P(E) < 1, 0 < P(F) < 1 and P(E|F) > P(E) . Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

- a. E and F are independent
- b. P(F | E^C) < P(F)
- c. P(F|E) > P(F)
- d. P(E | F^C) > P(E)

Question **6**

Complete

Marked out of 2.00

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

- [ ] a. 19/144
- [x] b. 13//144
- [ ] c. 125/432
- [ ] d. 325/432

Question **7**

Complete

Marked out of 2.00

The chances of failing in Physics are 20%

and in Mathematics are 10%

The chance of failing in at least one subject is

- [ ] a. 28%
- [ ] b. 62%
- [x] c. 38%
- [ ] d. 58%

Question **8**

Complete

Marked out of 2.00

A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

- a. pvalue is 0.97
- b. pvalue is 0.03
- c. pvalue is 0.02
- d. pvalue is 0.05

Question **9**

Complete

Marked out of 2.00

The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal. how much is critical region

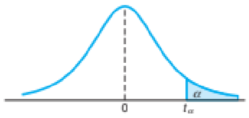
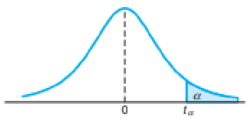


Table A.4 Critical Values of the t-Distribution

	α						
v	**0.40**	**0.30**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.538	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131



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3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.538	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131

- a. -1.79
- b. -1.69
- c. -1.39
- d. -1.56

Question **10**

Complete

Marked out of 2.00

The

_________of a test is the probability of rejecting H0 given that a specific alter-

native is true.

- a. strength
- b. tensor
- c. significance
- d. power

◄ CCN CSE

Jump to...

SMCS-CSE ►

Started on	Tuesday, 28 April 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 28 April 2026, 11:50 AM
Time taken	10 mins

Question 1
Complete
Marked out of 2.00

Find the wrong one from the given statements

- a. None of the above
- b. An n × n matrix with fewer than n distinct eigenvalues is not diagonalizable
- c. If A is diagonalizable, then transpose of A is also diagonalizable
- d. If A is diagonalizable and invertible, then A^{-1} is diagonalizable

Question 2
Complete
Marked out of 2.00

$$A = \begin{pmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{pmatrix}$$

If A is as given the for what values of a does the system AX=Y has unique solution for any Y

- a. 1
- b. -2
- c. None of the above
- d. Both

Question 3

Complete

Marked out of 2.00

Suppose that the characteristic polynomial of some matrix A is found to be

$p(\lambda) = (\lambda - 1)(\lambda - 3)^2 (\lambda - 4)^3$. What is the size of A?

- a. 5 X 5
- b. 5 X 6
- c. 6 X 5
- d. 6 X 6

Question 4

Complete

Marked out of 2.00

Let a,b,c be real numbers such that both b and c are not equal to 0.

Suppose P = [a b
c 0]

and P^{-1} = P. Then which of the following is true

- a. a = 0 and bc = -1
- b. a is not equal to 0 and bc = 1
- c. a = 0 and bc = 2
- d. a = 0 and bc =1

Question 5

Complete

Marked out of 2.00

Let E and F be two events with 0 < P(E) < 1, 0 < P(F) < 1 and P(E|F) > P(E) . Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

- a. P(F | E^C) < P(F)
- b. E and F are independent
- c. P(E | F^C) > P(E)
- d. P(F|E) > P(F)

Question C

Complete

Marked out of 2.00

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/U and P(F) < P(E). Then P(E) equals

- [ ] a. 3/4
- [x] b. 2/3
- [ ] c. 1/2
- [ ] d. 1/3

Question 7

Complete

Marked out of 2.00

The chances of failing in Physics are 20%

and in Mathematics are 10%

The chance of failing in at least one subject is

- [ ] a. 62%
- [ ] b. 38%
- [ ] c. 58%
- [x] d. 28%

Question 8

Complete

Marked out of 2.00

For comparing the means of weight in the two groups -smokers and non-smoker, which visualization is appropriate?

- (•) a. boxplot
- ( ) b. scatterplot
- ( ) c. pie chart
- ( ) d. line graph

Question 9

Complete

Marked out of 2.00

The

__________ of a test is the probability of rejecting H0 given that a specific alter-

native is true.

- a. significance
- b. tensor
- c. strength
- d. power

Question 10

Complete

Marked out of 2.00

The more prevalent situations involving tests on two means are those in which variances are unknown. If the scientist involved is willing to assume that both distributions are normal and that $\sigma 1 = \sigma 2 = \sigma$, the test used is

- a. one sample t-test
- b. marginal t-test
- c. variational t-test
- d. cross-origin t-test
- e. pooled t-test

◄ OS CSE

Jump to...

SMCS-CSE ►

Show one page at a time

Finish review

Question **1**

Complete

Marked out of 2.00

Flag question

If A is as given, then find sum of eigenvalues of A^{-1}

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

- a. 12/19
- b. None of the above
- c. 8
- d. 19/12

Question **2**

Complete

Marked out of 2.00

Flag question

$$A = \begin{pmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{pmatrix}$$

If A is as given the for what values of a does the system AX=Y has unique solution for any Y

- a. 1
- b. None of the above
- c. -2
- d. Both

Question **3**

Complete

Marked out of 2.00

Flag question

If A is as given, then A^{-1} can be given as

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

- a. $\frac{1}{4}((A^2+6A+9I))$
- b. $\frac{1}{4}((A^2-6A+9I))$
- c. $\frac{1}{4}((A^2-6A-9I))$
- d. None of the above

Question **4**

Complete

Marked out of 2.00

Flag question

Let P be a 4x4 matrix comprising of the following rows:

[1 2 0 2] , [-1 -2 1 1], [1 2 -3 -7], [1 2 -2 -4]

Then rank of P equals

- a. 2
- b. 3
- c. 1
- d. 4

Question **5**
Complete
Marked out of 2.00
Flag question

If X

is a continuous random variable with PDF f(x) = k x^2

for 0 <= x <= 1, the value of k

is:

- a. 4
- b. 1
- c. 3
- d. 2

Question **6**
Complete
Marked out of 2.00
Flag question

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

- a. 0.029
- b. 0.039
- c. 0.044
- d. 0.034

Question **7**
Complete
Marked out of 2.00
Flag question

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

- a. 325/432
- b. 13//144
- c. 125/432
- d. 19/144

Question **8**
Complete
Marked out of 2.00
Flag question

Blood Sample Data: In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given. Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes. Which test to use

- a. paired z test
- b. paired f test
- c. paired t test
- d. paired chisquare test

Question **9**
Complete
Marked out of 2.00
Flag question

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

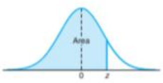

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110

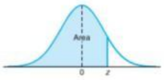

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

- a. +- 2.56
- b. +-2.96
- c. +-2.22

Question **10**

Complete

Marked out of 2.00

Flag question

An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. how many degree of freedom

- a. 18
- b. 20
- c. 10
- d. 15

Started on	Tuesday, 28 April 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 28 April 2026, 11:49 AM
Time taken	9 mins 51 secs

Question **1**

Complete

Marked out of 2.00

If A is as given, then A^{-1} can be given a

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

- a. $\frac{1}{4}((A^2 - 6A + 9I))$
- b. $\frac{1}{4}((A^2 + 6A + 9I))$
- c. $\frac{1}{4}((A^2 - 6A - 9I))$
- d. None of the above

Question **2**

Complete

Marked out of 2.00

Let a,b,c be real numbers such that both b and c are not equal to 0.

Suppose P = [a b
c 0]

and P^{-1} = P. Then which of the following is true

- [ ] a. a = 0 and bc = 2
- [x] b. a = 0 and bc =1
- [ ] c. a = 0 and bc = -1
- [ ] d. a is not equal to 0 and bc = 1

Question **3**

Complete

Marked out of 2.00

For two nonzero real numbers a and b, consider the system of linear equations :

ax + by = b/2

bx + ay = a/2

Which of the following statements is (are) TRUE?

- [ ] a. If both a+b and a-b are not equal to 0 , the system has no solution
- [x] b. If both a+b and a-b are not equal to 0 , the system has a unique solution
- [x] c. If a = -b , the solutions of the system lie on the line y-x = 1/2
- [x] d. If a = b , the solutions of the system lie on the line x + y =1/2

Question **4**

Complete

Marked out of 2.00

Let P be a 4x4 matrix comprising of the following rows:

[1 2 0 2] , [-1 -2 1 1], [1 2 -3 -7], [1 2 -2 -4]

Then rank of P equals

- [ ] a. 3
- [x] b. 4
- [ ] c. 2
- [ ] d. 1

Question **5**

Complete

Marked out of 2.00

If X

is a continuous random variable with PDF f(x) = k x^2

for 0 <= x <= 1, the value of k

is:

- [x] a. 3
- [ ] b. 4
- [ ] c. 1
- [ ] d. 2

Question **6**

Complete

Marked out of 2.00

If a variable X

follows a Poisson distribution such that P[X=1] = P[X=2]
then the mean of the distribution is:

- [ ] a. 3
- [ ] b. 4
- [x] c. 2
- [ ] d. 1

Question **7**

Complete

Marked out of 2.00

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/9 and P(F) < P(E). Then P(E) equals

- [ ] a. 1/2
- [x] b. 2/3
- [ ] c. 1/3
- [ ] d. 3/4

Question **8**

Complete

Marked out of 2.00

A __________ is the lowest level (of significance) at which the observed value of the test statistic is significant.

- a. z value
- b. alpha
- c. beta
- d. p value
- e. f value

Question **9**

Complete

Marked out of 2.00

A manufacturer of sports equipment has developed a new synthetic fishingline that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that μ = 8 kilograms against the alternative that μ is not equal to 8 kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance. What is critical region

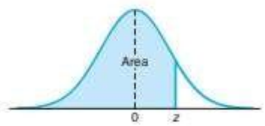

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

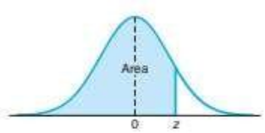

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

- a. +-2.96
- b. +-2.22
- c. +- 2.56

Question **9**

Complete

Marked out of 2.00

An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. how many degree of freedom

a. 20

b. 18

c. 10

d. 15

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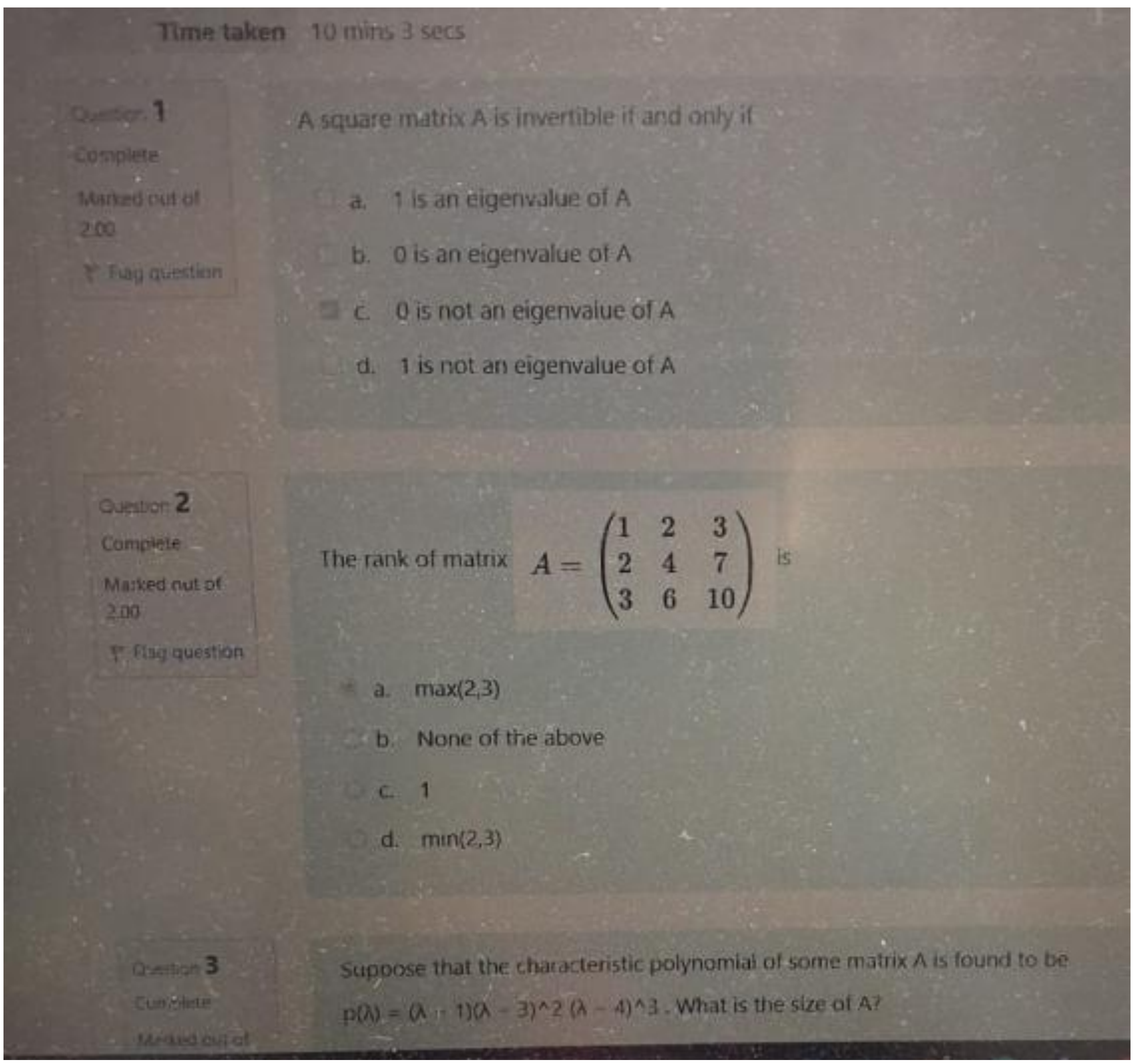

Time taken 10 mins 3 secs

Question 1

Complete

Marked out of 2.00

Flag question

A square matrix A is invertible if and only if

a. 1 is an eigenvalue of A

b. 0 is an eigenvalue of A

c. 0 is not an eigenvalue of A

d. 1 is not an eigenvalue of A

Question 2

Complete

Marked out of 2.00

Flag question

The rank of matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$ is

a. max(2,3)

b. None of the above

c. 1

d. min(2,3)

Question 3

Complete

Marked out of

Suppose that the characteristic polynomial of some matrix A is found to be p(λ) = (λ + 1)(λ − 3)^2 (λ − 4)^3 . What is the size of A?

Question **9**

Complete

Marked out of 2.00

uestion **3**

omplete

Marked out of
2.00

Flag question

Suppose that the characteristic polynomial of some matrix A is found to be

p(λ) = (λ − 1)(λ − 3)^2 (λ − 4)^3 . What is the size of A?

a. 5 X 6

b. 6 X 5

c. 5 X 5

d. 6 X 6

Question **4**

Complete

Marked out of
2.00

Flag question

If A is 3x3 matrix with all real entries, has an eigenvalue i

, then its other two eigenvalues can be

a. 0,-i

b. 2i,-2i

c. 0,1

d. -1,i

Question **5**

Complete

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

c. 38%

d. 58%

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/9 and P(F) < P(E). Then P(E) equals

a. 3/4

b. 1/3

c. 1/2

d. 2/3

8

Politicians are certainly interested in knowing what fraction of the voters will favor them in the next election, which test will be used for this purpose given to you a sample with voters and nonvoters.

a. f-test

b. chisquare test

c. z-test

d. t-test

9 The

Question **9**

Complete

Marked out of 2.00

Question 9

Complete

Marked out of 2.00

Flag question

The

________ of a test is the probability of rejecting H0 given that a specific alter-

native is true.

a. tensor

b. significance

c. strength

d. power

Question 10

Complete

Marked out of 2.00

Flag question

A __________ is the lowest level (of significance) at which the observed value of the test statistic is significant.

a. p value

b. z value

c. f value

d. beta

e. alpha

Question **9**

Complete

Marked out of 2.00

Question 1

Complete
Marked out of 2.00

Find the wrong one from the given statements

Answer

a. If A is diagonalizable, then transpose of A is also diagonalizable ☐

b. None of the above ☐

c. If A is diagonalizable and invertible, then A^{-1} is diagonalizable ☐

d. An n × n matrix with fewer than n distinct eigenvalues is not diagonalizable ☑

‹ Show all questions on one page ›

Question **9**

Complete

Marked out of 2.00

Question 2

Complete
Marked out of 2.00

The rank of matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$ is

Answer

- [x] a. max(2,3)
- [ ] b. min(2,3)
- [ ] c. 1
- [ ] d. None of the above

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Question 3

Complete
Marked out of 2.00

If A is 3x3 matrix with all real entries, has an eigenvalue i
, then its other two eigenvalues can be
Answer

a. 0,1

b. 0,-i

c. -1,i

d. 2i,-2i

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Question 4

Complete
Marked out of 2.00

Let P be a 4x4 matrix comprising of the following rows:

[1 2 0 2] , [-1 -2 1 1], [1 2 -3 -7], [1 2 -2 -4]

Then rank of P equals

Answer

- a. 3 ☐
- b. 4 ☐
- c. 2 ☑
- d. 1 ☐

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Question 5

Complete
Marked out of 2.00

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/9 and P(F) < P(E). Then P(E) equals

Answer

a. 1/2 ☐

b. 3/4 ☑

c. 1/3 ☐

d. 2/3 ☐

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Question 6

Complete
Marked out of 2.00

If X
is a continuous random variable with PDF f(x) = k x^2

for 0 <= x <= 1, the value of k

is:

Answer

a. 4

b. 3

c. 2

d. 1

Show all questions on one page

6 of 10

Question **9**

Complete

Marked out of 2.00

9 mins 43 secs

Question 7

Complete
Marked out of 2.00

Let E and F be two events with 0 < P(E) < 1, 0 < P(F) < 1 and P(E|F) > P(E) . Let E^C and F^C denote the complement of E and F respectively. Which of the following statements is (are) TRUE?

Answer

- [ ] a. P(E | F^C) > P(E)
- [x] b. E and F are independent
- [ ] c. P(F | E^C) < P(F)
- [x] d. P(F|E) > P(F)

< Show all questions on one page >

Question **9**

Complete

Marked out of 2.00

Question 8

Complete
Marked out of 2.00

For comparing the means of weight in the two groups -smokers and non-smoker, which visualization is appropriate?

Answer

a. pie chart ○

b. boxplot ○

c. line graph ○

d. scatterplot ◉

< Show all questions on one page >

Question **9**

Complete

Marked out of 2.00

Question 9

Complete

Marked out of 2.00

The

_________ of a test is the probability of rejecting H0 given that a specific alter-
native is true.

Answer

a. strength

b. significance

c. power

d. tensor

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Question 10

Complete
Marked out of 2.00

Blood Sample Data: In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given. Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes. Which test to use

Answer

a. paired chisquare test

b. paired t test

Show all questions on one page

Question **9**

Complete

Marked out of 2.00

Dev
28/4/2026 at 20:29

Question **5**
Complete
Marked out of 2.00
Flag question

The probability of obtaining at least two sixes when a fair die is thrown 4 times is:

- [ ] a. 19/144
- [ ] b. 125/432
- [ ] c. 13//144
- [x] d. 325/432

Question **6**
Complete
Marked out of 2.00
Flag question

The total number of accidents in a factory per month follows a Poisson distribution with a mean of 5.2. The probability that less than 2 accidents occur in a month is:

- [x] a. 0.039
- [ ] b. 0.044
- [ ] c. 0.029
- [ ] d. 0.034

Dev
28/4/2026 at 20:29

Let E and F be two independent events with P(E|F) + P(F|E) = 1, P(E intersection F) = 2/9 and P(F) < P(E). Then P(E) equals

- [ ] a. 3/4
- [ ] b. 1/2
- [ ] c. 2/3
- [x] d. 1/3

The more prevalent situations involving tests on two means are those in which variances are unknown. If the scientist involved is willing to assume that both distributions are normal and that $\sigma 1 = \sigma 2 = \sigma$, the test used is

- a. cross-origin t-test
- b. pooled t-test
- c. one sample t-test (selected)
- d. variational t-test
- e. marginal t-test

Question **9**

Complete

Marked out of 2.00

Dev
28/4/2026 at 20:29

Politicians are certainly interested in knowing what fraction of the voters will favor them in the next election. which test will be used for this purpose given to you a sample with voters and nonvoters.

- a. f-test
- b. chisquare test
- c. z-test
- d. t-test

The

________ of a test is the probability of rejecting H0 given that a specific alter-

native is true.

- a. tensor
- b. power
- c. significance
- d. strength